

Repo Audit & Reorganization

Prompted by a real GitHub Actions failure email, not a scheduled review — what the audit found, the folder restructure that came out of it, and every reference that had to be updated to keep it working. Standalone, not tied to a numbered build.

STATUS

Complete — repo reorganized, CI re-verified green

TRIGGER

A CI failure email after the first push of Build 6 Phase 2 onward

FOLLOWS

Build 7 (verifier/critic agent)

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What Triggered This

Not a scheduled review — a GitHub Actions failure email arrived after a routine push (stripping a co-author trailer from git history). Checking it against real run data surfaced something bigger than the immediate failure: `origin/main` had been 23 commits behind local for a while. All of Build 6 Phase 2 onward, and all of Build 7, had been built and committed locally but never pushed — that push was the first time any of it reached CI at all.

The lesson generalizes past this one incident. A green CI badge only means something if CI actually ran against the code being described. Work sitting local-only, however well-tested by hand, isn't verified by a CI pipeline that's never seen it.

The CI Bug Itself

CI correctly went red. 25 of 27 tests passed; `test_chat_rounds.py`'s two DB-touching tests failed with relation `"agent_scratch.chat_rounds"` does not exist.

Root cause: `ci.yml`'s migration step is a hardcoded file list. `012_chat_rounds_schema.sql` (Build 6 Phase 3.5, persistent chat history) was added to the repo but never added to that list — in either the `test` job or the `eval` job. Same bug class as `extract_doc_chunks.py`'s old hand-maintained `SOURCE_FILES` list, which had already caused two separate incidents earlier in the project before being fixed with directory-scan auto-discovery.

Fixed: both migration lists now include `012_chat_rounds_schema.sql`. Confirmed via a real `gh run watch` to a fully green run, not assumed from the diff alone.

Full Audit Findings

The CI bug prompted a wider question: what else has drifted the same way?

Checked every hardcoded list, generated-vs-real-file relationship, and config-vs-code dependency in the project.

Build 7's verifier eval harness never ran in CI.

`run_verify_eval.py/verify_eval_cases.py` weren't invoked anywhere in `ci.yml`, and `verify_answer.py` wasn't in the `detect-changes` paths-filter — a change to the verifier's own logic wouldn't even re-trigger the eval job. Fixed: both added.

.env.example documented 4 of the 10 env vars the code actually requires.

`ANTHROPIC_API_KEY` and both reader/writer role credentials were entirely absent. `tools.py` reads most of these with no default (`os.environ[...]`), so a fresh clone following the file as written hits an immediate `KeyError`. Rewritten with all 10, each with a one-line note on why it's needed.

No README.md existed.

A repo whose entire purpose is a portfolio artifact had no entry point for a visitor landing on it. Added.

db_test_py.py — a stray early debugging script.

Same spirit as Build 5's deleted `api_test.py`: an early Postgres connectivity check querying a `customers` table no longer in the agent's actual scope. Its filename doesn't match either of pytest's discovery globs, so it was never at risk of silent collection — but it was dead, untested code sitting in the project root. Deleted.

An empty VS Code scratch file was tracked in git.

Local `Postgres.session.sql`, 0 bytes, auto-created by a SQL client extension. Deleted.

Two vestigial SQL Server files, unreferenced by anything.

`schema_mssql.sql/seed_mssql.sql` — part of the original dual-database lab environment, not connected to any code, CI step, or Docker config in the actual seven-build narrative. Relocated to `legacy/` rather than deleted, since `docker-compose.yml`'s `mssql_local` service still exists.

Verified clean, not assumed: `Dockerfile`'s `COPY` list against the actual file set, `requirements.txt` against every top-level import actually used, `.gitignore` coverage (including confirming `.venv/` was never at risk — it has its own internal `.gitignore`), and `extract_doc_chunks.py`'s glob-based doc discovery (already fixed correctly in an earlier build, confirmed not regressed).

The Reorganization

A flat ~50-file repo root had never been restructured as the project grew across seven builds. Before moving anything, the actual Python import graph was checked directly — `run_eval.py/run_verify_eval.py` import `agent/verify_answer` directly, and every test file imports `tools/agent/logging_utils` directly. That ruled out splitting "core app" from "pipeline/eval scripts" into separate directories without adding `sys.path` configuration everywhere — so every first-party `.py` file stays together, flat, in one directory.

FOLDER	CONTENTS
<code>src/</code>	All 22 first-party Python modules and scripts, flat — required by the import graph above.
<code>migrations/</code>	The 12 numbered SQL migrations, unchanged content.
<code>tests/</code>	All 7 <code>test_*.py</code> files. <code>pytest.ini</code> gained <code>testpaths = tests</code> and <code>pythonpath = src</code> so <code>import tools</code> etc. keeps resolving with zero changes to the test files themselves.
<code>docs/</code>	Renamed from <code>PROJECT_DIAGRAMS/</code> . Every per-build diagram/brief folder, plus the cheat sheet, moved in.
<code>legacy/</code>	The two vestigial SQL Server files.
<code>data/</code>	Unchanged location — already its own folder.

Every reference updated, not just the files themselves: `Dockerfile` (`COPY/CMD` now target `src/`), `docker-compose.yml` (`streamlit_app`'s command), `ci.yml` (both migration-loop path prefixes, every `python script.py` invocation, the `paths-filter` list), `extract_doc_chunks.py` (its `glob` target and hardcoded seed-file paths, since it reads `docs/` by relative path from the repo root regardless of its own new location in `src/`), and `.gitignore` (the personal working-style folder's path).

All moves done via `git mv`, not delete-and-recreate. Git records each as a rename, preserving `git log --follow` history for every file rather than starting it over.

One known follow-up, not yet done: `pg_local`'s `docs.chunks` table still has rows keyed by the old `PROJECT_DIAGRAMS/...` source-file strings. The next time `extract_doc_chunks.py` runs locally, those become orphaned duplicates alongside the new `docs/...`-keyed rows — same failure shape as the backslash/forward-slash duplicate bug from Build 6, different cause. Fix when next touching the docs-RAG pipeline locally: `DELETE FROM docs.chunks WHERE source_file LIKE 'PROJECT_DIAGRAMS/%' OR source_file = 'sql-test-01-cheatsheet.html'` before re-running the extract/embed pair. CI is unaffected — it always starts from an empty database.

Verification

py_compile across every moved file

Clean on all 22 `src/` files and all 7 `tests/` files.

Full Docker image build

Built against the reorganized paths end to end before committing, not assumed from reading the Dockerfile diff.

Live CI run, watched to completion

`detect-changes` ✓, `test` ✓ (all 27), `eval` correctly gated — confirmed via `gh run watch`, the same live-verification standard every prior build in this project has used.

Cheat sheet re-verified both ways

`--screenshot` and `--print-to-pdf/pdftoppm`, after every path reference across the Project Files table was rewritten to match the new structure.

Repo audit and reorganization complete — CI green against the real, current file structure for the first time since Build 6 Phase 2. Ongoing cross-build command reference lives in `docs/sql-test-01-cheatsheet.html`.